Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Keep mass fixed and multiply speed by 3. By what factor does kinetic energy change? Explain from the equation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

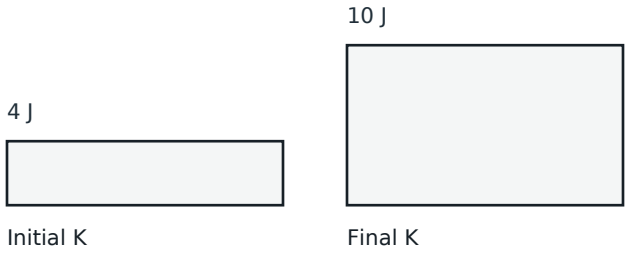
Worked reasoning: A 2 kg cart at 2 m/s has K=4 J. Adding 5 J of net work gives final K=9 J, so final speed is 3 m/s.

Your turn: Draw before-and-after energy bars and label the transfer between them.

Marzano check - circle: ① ② ③

03 Look and explain

Energy bars show kinetic energy changing from 7 J to 10 J. Find net work and describe the speed change for constant mass.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart moves at 3 m/s. Find $K=\frac{1}{2}\times2\times3^2$. If net work adds 7 J, find final kinetic energy.

Marzano check - circle: ① ② ③

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05 Use the idea

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Marzano check - circle: 1 2 3

06 Choose a method

Two 4 kg carts latch together after colliding. Is this perfectly inelastic? State what remains conserved if external impulse is negligible.

Marzano check - circle: 1 2 3

07 Exit check

K changes from 20 J to 8 J. Find net work and state whether speed rises or falls.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Keep the cart's mass fixed. Compare the energy needed to increase speed from 1 to 2 m/s with that from 2 to 3 m/s. Use m=2 kg.

Marzano check - circle: 1 2 3